Murat Bilgehan Ertan

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*I work towards **safe**, **secure**, and **private** AI that remains practically useful.*

Education

Centrum Wiskunde & Informatica (CWI) - Vrije Universiteit Amsterdam **May 2025 – Apr 2029**
PhD Student in AI Safety & Privacy *Amsterdam, Netherlands*

- PhD student under Prof. Marten van Dijk on AI safety and privacy-preserving machine learning, combining theoretical analysis with practical engineering. Currently collaborating with Prof. Srini Devadas (MIT) on certified defenses against data poisoning in LLM post-training and on the safety of agentic systems. First-author papers at NeurIPS 2026 (PAC-private fine-tuning of LLMs) and ACM CCS 2026 (fundamental limits of privacy–utility trade-offs in DP-SGD). Building and benchmarking safe and secure training pipelines and agentic systems.

Vrije Universiteit Amsterdam **Sep 2023 - Mar 2025**
MSc in Computer Security - Cum laude distinction *Amsterdam, Netherlands. cGPA: 9.1/10.0*

- Completed the two-year master's program in **1.5 years**, finishing six months ahead of schedule with a perfect thesis grade (10/10) and with a cumulative GPA of 9.1/10.
- **Master's Thesis:** *ROAR: Robust Object Removal and Re-annotation*: proposed a scalable framework for privacy-preserving dataset obfuscation that removes sensitive objects using instance segmentation and generative inpainting, achieving strong privacy guarantees with minimal loss in detection and reconstruction performance. Supervised by Prof. Marten van Dijk.
- **Teaching Assistant (Nov'23 – Jan'24):** Assisted in the Secure Programming course, guiding weekly labs and discussions for a C++-based secure chat application project focused on applied cryptography and secure coding practices.
- **Key Courses:** Software Security (8.5/10), Network Security (8.5/10), Security and Machine Learning (10/10), Binary and Malware Analysis (8/10), Hardware Security (10/10), Advanced Operating Systems (8/10).

Sabancı University **Sep 2019 – June 2023**
BSc in Computer Science and Engineering *Istanbul, Turkey. cGPA: 3.59/4.00*

- **Key Courses:** Computer and Network Security, Artificial Intelligence, Machine Learning, Cybersecurity Practices and Applications
- **Honors:** Dean's List - High Honor (5 Semesters), Honor (1 Semester).
- Co-founded and co-led SUCyber, the university's cybersecurity student association, for three years. Designed and delivered an 8-week workshop series each semester covering Linux, network security, web exploitation, reverse engineering, and forensics.
- Volunteered in nursing homes and mentored first-year students through the Civil Involvement Project (CIP); board member of the Mimar ve Mühendisler Grubu (MMG) NGO, supporting educational outreach and student–industry collaboration.

Work Experience

PRODAFT **Aug 2022 – May 2025**
Security Researcher & Developer *Netherlands, EC3 partner of EUROPOL*

- Worked on developing production-grade frameworks and pipelines both AI-driven and traditional for automated threat intelligence and security analysis. Utilized mostly C++ and Python to build scalable systems, integrating machine learning and LLM-based components to enhance detection, reasoning, and analyst support. One project led to a peer-reviewed publication on behavior-based threat actor attribution in *ACM DTRAP* 2024. [Paper Link].

KocLab **Mar 2022 – Feb 2023**
Research Assistant *Istanbul, Turkey*

- Worked under Prof. Çetin Kaya Koç on projects involving Reinforcement Learning and Cyber-Physical Security. Applied reinforcement learning to the safe navigation of autonomous drones, designing experimental setups in PEDRA.

Turkcell **Jun 2021 – Aug 2021**
Security Management & Architecture Intern *Istanbul, Turkey*

- Researched HTTP Secure Headers, Password Policies, Session Management, and JWT Token Security under the supervision of Necati Ersen Siseci (Offensive Security Associate Director).

Publications

Mahmood, K., **M. B. Ertan** and van Dijk, M. "Beyond Stop Signs: Why Evasion Attacks Matter Even More" *IEEE Security & Privacy, 24(3):65–70, 2026.* [IEEE Xplore].

M. B. Ertan, Zhu, X., Nguyen, P. H., van Dijk, M. and Devadas, S. "PACZero: PAC-Private Fine-Tuning of Language Models via Sign Quantization" *Accepted at NeurIPS'26 (poster).* [arXiv preprint].

Marten van Dijk and **M. B. Ertan**. "Trade-off Functions for DP-SGD with Subsampling based on Random Shuffling: Tight Upper and Lower Bounds" *Under submission; 2026.* [arXiv preprint].

M. B. Ertan, and Marten van Dijk "Fundamental Limitations of Favorable Privacy–Utility Guarantees for DP-SGD" *Accepted ACM Conference on Computer and Communications Security (CCS)*, 2026. [arXiv preprint].

M. B. Ertan, Böge, E., Chen, M., Mahmood, K. and van Dijk, M. "On the Evidentiary Limits of Membership Inference for Copyright Auditing" *Under submission; 2026.* [arXiv preprint].

M. Damie, **M. B. Ertan**, D. Essoussi, A. Makhanu, G. Peter, R. Wensveen. "TOSSS: A CVE-based Software Security Benchmark for Large Language Models." *Proceedings of the 12th ACM International Workshop on Security and Privacy Analytics (IWSPA '26), pp. 111–117, 2026.* [arXiv preprint] [ACM Library].

E. Böge, Y. Günindi, **M. B. Ertan**, E. Aptoula, N. Alp, H. Özkan. "A Biologically Inspired Filter Significance Assessment Method for Model Explanation." *The 3rd World Conference on eXplainable Artificial Intelligence (xAI-2025)*, Istanbul, Turkey, July 9–11, 2025. *Communications in Computer and Information Science (CCIS)*, Springer Nature, November 2025. [Springer Link]. Presented at the conference; slides available at [Sabancı University Research Repository].

M. B. Ertan, Sahu, R., Nguyen, P.H., Mahmood, K. and van Dijk, M. "Beyond Anonymization: Object Scrubbing for Privacy-Preserving 2D and 3D Vision." *Master's Thesis, Vrije Universiteit Amsterdam*, March 2025. [arXiv preprint]. Awarded final grade: 10/10.

M. B. Ertan, E. Böge (equal contribution), H. Alptekin, O. Çetin. "Unveiling Cyber Threat Actors: A Hybrid Deep Learning Approach for Behavior-based Attribution." *ACM Digital Threats: Research and Practice (DTRAP)*, Accepted June 2024. [Link to paper].

Grants, Programs & Activities

Course Instructor – AI Agents in Practice (CWI Amsterdam) **Nov 2026 – Jan 2027**

- Will teach *AI Agents in Practice* (co-instructor: Prof. Marten van Dijk), a five-session hands-on course for CWI researchers on agent fundamentals, agentic workflows, LLM fundamentals, and safety.

EuroHPC JU Benchmark Access Grant – JUPITER Booster **Sep 2026**

- Wrote the full proposal *"Scalable Foundations for Safe and Certifiable Post-Training of Large Language Models"* (EHPC-BEN-2026B09-161; PI: Prof. Marten van Dijk) as its sole author. Awarded **2,000 node hours** (≈8,000 GPU hours on NVIDIA GH200) on JUPITER, Europe's first exascale supercomputer, for a 3-month allocation.

Peer Review **2026**

- Reviewer for *IEEE Transactions on Emerging Topics in Computing* (TETC). Review record on Web of Science.

Guest Lecture – Data Protection Technologies (MSc, University of Amsterdam) **Mar 2026**

- Invited lecture (coordinator: Dr. Francesco Regazzoni) in the UvA/VU MSc core course on data protection, presenting my CCS 2026 accepted paper.

Summer Schools **2025 – 2026**

- *Attended:* Machine Learning Summer School (MLSS) 2026, Max Planck Institute for Intelligent Systems, Tübingen; ICAI Summer School 2026, ICAI (UvA & VU), Amsterdam; and the CUSO Workshop on Trustworthy Generative AI, University of Neuchâtel (2025).

Young Talents in Cybersecurity 2025 (FR–NL Bilateral Program) **Mar 2025 – Oct 2025**

- Selected as one of 12 young talents from France and the Netherlands; co-developed TOSSS, a benchmark for LLM poisoning in secure code generation, awarded first place in the 2025 challenge, and presented it at Forum InCyber FIC (Lille), ONE Conference (The Hague), and the MIT CAMS Research Discussion Group.

Open-Source Contributions – Hugging Face **2024**

- Merged PRs to `huggingface_hub` (user API endpoints [#2147]; CLI repository tag management [#2172]) and `lighteval` (restored the BigBench Hard task suite [#126]).

Technical Skills

Programming Languages: Python, C, Node.JS, Assembly.
Languages: Turkish (Native), English (Fluent), Dutch (A2).
Tools/Frameworks: PyTorch & Opacus, JAX & JAX Privacy, Hugging Face, TensorFlow, Ghidra, Slurm (HPC).
Interests: Differential Privacy, Privacy-Preserving ML, AI Safety, Deep Learning, Cryptography, Software Security.